Lecture 2

Lecture 2 develops the BTZ black hole and its thermodynamics, the Fefferman–Graham expansion (which *terminates* in three dimensions), the resulting Bañados phase space and its supersymmetric loci, the Brown–Henneaux asymptotic symmetry analysis giving $c = 3\ell/2G$, and — more briefly — the Chern–Simons reformulation and the Liouville/boundary-graviton description of the asymptotic dynamics.

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Conventions

Signature $(-,+,+)$; $\mu,\nu = 0,1,2$; $R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu \Gamma^{\rho}_{\nu\sigma} - \ldots$, $R_{\mu\nu} = R^{\rho}{}_{\mu\rho\nu}$, so that the round sphere has $R > 0$. Units $c = \hbar = 1$. The action is

$$I = \frac{1}{16\pi G}\int \mathrm{d}^3x\,\sqrt{-g}\,\big(R - 2\Lambda\big) + (\text{boundary terms}), \qquad \Lambda = -\frac{1}{\ell^2} \quad (\Lambda < 0 \text{ case}). \tag{1}$$

The dimension of G follows from the action: $[\mathrm{d}^3x] = L^3$ and $[R] = L^{-2}$, so $\frac{1}{G}\int \mathrm{d}^3x\sqrt{-g}\,R$ is dimensionless only if

$$[G] = \text{length} = \frac{1}{\text{mass}} \qquad (\hbar = c = 1), \tag{2}$$

so that GM is *dimensionless*. Contrast four dimensions, where $[G_4] = \text{length/mass}$ and $G_4 M$ is the Schwarzschild radius: in $3d$ no length can be built from G and M alone, and masses are naturally measured in units of $1/G$. Two consistency checks on (2): the central charge $c = 3\ell/2G$ of §3 and the Chern–Simons level $k = \ell/4G$ of §4 are pure numbers, which forces $[G] = [\ell]$. Also dimensionless are GJ/ℓ, $M\ell$, $\mathcal{L}$, and the entropy $S = \pi r_+/2G$. The Planck length appears only through $\ell_{\mathrm{P}} = \hbar G$.

Some references instead set $8G = 1$ and treat M as dimensionless; then G is a pure number but c and k carry dimensions. We do not use that convention.

Boundary lightcone coordinates are

$$x^{\pm} = \frac{t}{\ell} \pm \varphi, \qquad \varphi \simeq \varphi + 2\pi, \qquad \partial_{\pm} = \tfrac{1}{2}\Big(\ell\,\partial_t \pm \partial_\varphi\Big). \tag{3}$$

1 The BTZ black hole

1.1 The metric

The hyperbolic–hyperbolic quotient of §2.2 (lecture 1), written in adapted coordinates, is the BTZ black hole [11, 12]:

$$\boxed{\mathrm{d}s^2 = -f(r)\,\mathrm{d}t^2 + \frac{\mathrm{d}r^2}{f(r)} + r^2\Big(\mathrm{d}\varphi + N^{\varphi}(r)\,\mathrm{d}t\Big)^2} \tag{4}$$

with

$$f(r) = -8GM + \frac{r^2}{\ell^2} + \frac{16G^2J^2}{r^2} = \frac{(r^2 - r_+^2)(r^2 - r_-^2)}{\ell^2 r^2}, \qquad N^{\varphi} = -\frac{4GJ}{r^2} = -\frac{r_+ r_-}{\ell\, r^2}. \tag{5}$$

[checked symbolically] The two roots of f are

$$r_\pm^2 = 4G\ell^2\Big[\,M \pm \sqrt{M^2 - J^2/\ell^2}\,\Big], \qquad \Longleftrightarrow \qquad M = \frac{r_+^2 + r_-^2}{8G\ell^2}, \quad J = \frac{r_+ r_-}{4G\ell}. \tag{6}$$

The roots (6) are real and non-negative — equivalently, f actually vanishes somewhere on $r > 0$, so that a horizon exists at all — precisely when

$$\boxed{M \geq 0, \qquad |J| \leq M\ell.} \tag{7}$$

To see where this comes from, minimise f over $r > 0$: the stationary point sits at $r_*^2 = 4G|J|\ell$ and

$$f_{\min} = f(r_*) = \frac{8G}{\ell}\big(|J| - M\ell\big) \qquad \text{[checked]}, \tag{8}$$

so $f_{\min} \le 0 \iff M\ell \ge |J|$ (which already forces $M \ge 0$). Note that (7) is a *definition of the black-hole branch*, not a smoothness or regularity condition: nothing is singular when it fails. The outer horizon r_+ is the event horizon; r_- is a Cauchy horizon. There is also an ergosphere: $g_{tt} = 0$ at $r_{\rm erg}^2 = 8GM\ell^2 = r_+^2 + r_-^2$, and ∂_t is spacelike for $r_+ < r < r_{\rm erg}$.

Remark 1 (What (7) does and does not exclude)**.** Condition (7) is closely related to, but not the same as, the requirement that closed timelike curves be hidden. Since $g_{\varphi\varphi} = r^2$ (§2.3 (lecture 1)), the chronology-violating region $r^2 < 0$ is present for *every* member of the family with $J \neq 0$; what varies is whether it is hidden. Three cases:

(a) $M \ge 0$, $|J| \le M\ell$ — **black holes.** Here $0 < r_- \le r_+$ for $J \neq 0$, so the surface $r = 0$ and the CTCs behind it lie inside the inner horizon. The exterior is protected. This is the branch (7) defines.

(b) $|J| > M\ell$ — **overspinning, excluded.** By (8) $f > 0$ everywhere, so there is no horizon, and $r = 0$ is a *naked* timelike boundary visible from the conformal boundary. This is a cosmic-censorship exclusion, and it is why the wedge $|J| > M\ell$ is hatched out of Fig. 3. In the (g_L, g_R) classification of §2.2 (lecture 1) it is the *mixed* hyperbolic × elliptic class — the one case absent from that table, being neither a black hole nor a point particle.

(c) $M < 0$, $|J| \le |M|\ell$ — **conical defects, retained.** These fail (7) too, and $f_{\min} > 0$ automatically, so they have no horizon either — yet we keep them, because they are point particles rather than overspun black holes. Note that the *spinning* defect is not innocent: writing $r_\pm = i\ell a, i\ell b$ gives $f = (r^2 + \ell^2 a^2)(r^2 + \ell^2 b^2)/\ell^2 r^2 \to +\infty$ as $r \to 0$, so $r = 0$ is timelike and at finite proper distance ($\sim \epsilon^2/2ab\ell$) [checked], and its CTC region is naked in exactly the same way as in case (b). This is the AdS version of the spinning point particle of §1.3 (lecture 1), and it is tolerated on the usual grounds: a real rotating source is extended, and the CTC region would lie inside it. Only the *static* defect is genuinely clean — there $f = \alpha^2 + r^2/\ell^2$ and $r = 0$ is a regular conical tip, not a boundary.

So: (7) $\iff$ "a horizon exists". Chronology protection is a *separate* statement, and by itself it would not draw the same line.

Remark 2 (The $\Lambda \to 0$ obstruction)**.** Note that $f(r)$ has no horizon if $\ell \to \infty$: the r^2/ℓ^2 term is what makes f grow at large r and turn over. This is the technical reason there is no asymptotically flat $3d$ black hole — consistent with §1.4 (lecture 1), where increasing the mass closed the universe rather than forming a horizon.

1.2 No curvature singularity; CTCs behind the inner horizon

By (5) (lecture 1) the BTZ metric has

$$R_{\mu\nu\rho\sigma} = -\frac{1}{\ell^2}\big(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho}\big), \qquad R = -\frac{6}{\ell^2}, \qquad R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = \frac{12}{\ell^4}, \tag{9}$$

everywhere, for *all* M, J. [checked symbolically] There is no curvature singularity: BTZ is a smooth quotient of AdS_3. What, then, is the "singularity" at $r = 0$?

The identification Killing vector is $\xi = \partial_\varphi$, with

$$\xi \cdot \xi = g_{\varphi\varphi} = r^2. \tag{10}$$

Continuing the coordinate r^2 to negative values, ξ becomes timelike and its orbits — which are *closed* by construction — are closed timelike curves. Following BHTZ [12] one excises $r^2 < 0$; the surface $r = 0$ is then a boundary of the spacetime, a **singularity in the causal structure** rather than in the geometry. For $J \neq 0$ the region $r^2 < 0$ is causally disconnected from the exterior by the inner horizon r_-, so the exterior region $r > r_+$ (and indeed $r > r_-$) is perfectly well behaved.

For $J = 0$ the two horizons merge at $r_- = 0$ and the CTC region touches the would-be singularity; for $|J| = M\ell$ (extremal) $r_+ = r_-$ and the two horizons coincide. The conformal (Penrose) diagrams are drawn in Fig. 1. Structurally they are those of Schwarzschild–AdS (non-rotating), Reissner–Nordström–AdS (non-extremal rotating) and extremal Reissner–Nordström (extremal), with two modifications that are specific to $3d$:

- what is a curvature singularity in the $4d$ cases is here only the causal boundary $r = 0$ of §3.2 — the curvature is $R = -6/\ell^2$ right up to it;
- the timelike lines marked $r = \infty$ are the AdS conformal boundary, where the dual CFT lives; their presence is what makes the notion of a black hole horizon meaningful at all (cf. §2.4 (lecture 1)).

Exercise 1. Show that $r = 0$ for $J \neq 0$ is a *timelike* surface and that the BTZ black hole is therefore not globally hyperbolic once the region $r^2 < 0$ is excised. Compare with the Kerr inner horizon.

1.3 Mass and angular momentum

M and J as defined in (6) are not put in by hand: they are the canonical (Regge–Teitelboim [7]) charges generating ∂_t and $-\partial_\varphi$. In the ADM decomposition the on-shell variation of the Hamiltonian is a pure boundary term,

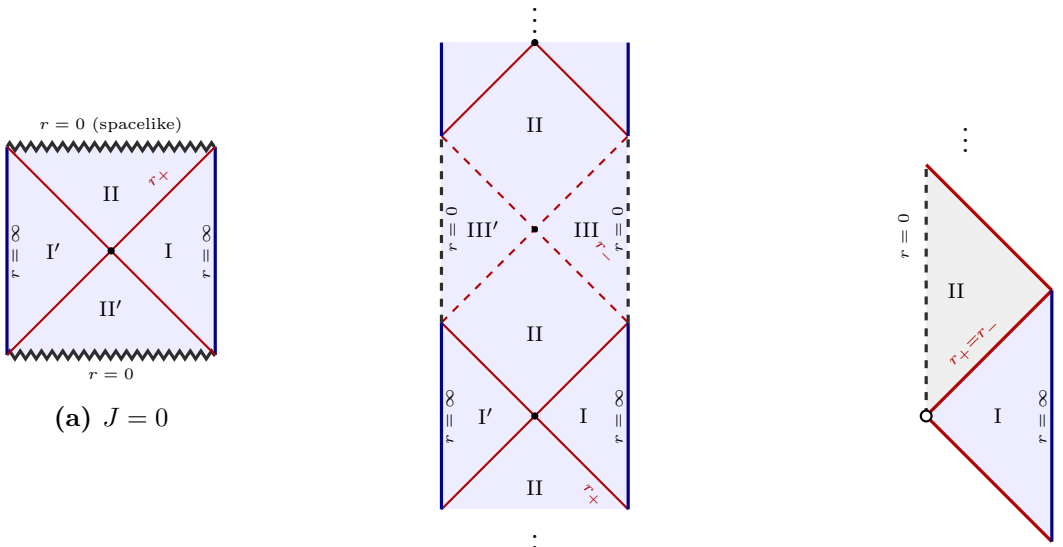
$$\delta H[\zeta] = \frac{1}{16\pi G} \oint_{r\to\infty} \mathrm{d}S_i \left[G^{ijkl}\big(\zeta^\perp \nabla_j \delta h_{kl} - \partial_j \zeta^\perp\, \delta h_{kl}\big) + 2\zeta_k\, \delta\pi^{ik} \right]. \tag{11}$$

Here h_{ij} is the metric induced on the constant-t slice, π^{ij} its conjugate momentum, ∇ the Levi-Civita connection of h, and $\zeta^\perp, \zeta^k$ the normal and tangential parts of ζ (lapse and shift). The surface element is $\mathrm{d}S_i = n_i \sqrt{\sigma}\, \mathrm{d}\varphi$, with n_i the outward unit normal of the circle at large r and $\sqrt{\sigma}$ its induced measure. Finally G^{ijkl} is the **DeWitt supermetric**,

$$\boxed{G^{ijkl} = \tfrac{1}{2}\big(h^{ik}h^{jl} + h^{il}h^{jk}\big) - h^{ij}h^{kl} \;=\; \tfrac{1}{2}\big(h^{ik}h^{jl} + h^{il}h^{jk} - 2h^{ij}h^{kl}\big).} \tag{12}$$

It is not an extra ingredient: it is exactly the tensor that multiplies $\nabla\nabla\,\delta h$ in the boundary term generated by varying the curvature density, $\sqrt{h}\, h^{ij}\delta R_{ij} = \nabla_i\big[\sqrt{h}\, G^{ijkl}\nabla_j \delta h_{kl}\big]$, which is why it appears here and nowhere else.

Two properties of (12). *(1) The relative factor of 2 is fixed, not conventional.* Repeating the static-BTZ computation below with $G^{ijkl} = \frac{1}{2}(h^{ik}h^{jl} + h^{il}h^{jk} - 2c\, h^{ij}h^{kl})$ gives an integrand independent of r *only* for $c = 1$; for $c = \frac{1}{2}$ or $c = 0$ it retains r-dependence [checked]. *(2) In three dimensions it is not an inverse.* In d spatial dimensions the inverse of $G_{ijkl} = \frac{1}{2}(h_{ik}h_{jl} + h_{il}h_{jk} - h_{ij}h_{kl})$ carries coefficient $2/(d-2)$, which reproduces (12) for $d = 3$ but diverges for $d = 2$: acting on symmetric tensors, $(G\,S)_{ij} = S_{ij} - \frac{1}{2}h_{ij}\,\mathrm{tr}\,S$ annihilates $S_{ij} = h_{ij}$ precisely when $d = 2$, so on our spatial slices the DeWitt metric is *degenerate* and has no inverse [checked], so (12) stands as a definition in its own right.



(a) $J = 0$ **(b)** $0 < |J| < M\ell$ **(c)** $|J| = M\ell$

Figure 1: Conformal diagrams of the BTZ black hole; compare Carlip's review [15], whose Fig. 1b is our (a), Fig. 1a is our (b) and Fig. 1c is the extremal case. Blue: the timelike AdS conformal boundary $r = \infty$. Red: horizons (solid r_+, dashed r_-). Regions are labelled as in [15]: I is $r \geq r_+$, II is $r_- \leq r \leq r_+$, III is $0 < r < r_-$. **(a)** $J = 0$: $f(0) = -r_+^2/\ell^2 < 0$, so $\partial_\mu r$ is timelike and $r = 0$ is a *spacelike* boundary (grey zigzag) — the diagram is Schwarzschild-like, "similar in structure, except for its asymptotic behaviour, to the diagram for the ordinary Schwarzschild solution" [15]. **(b)** $0 < |J| < M\ell$: $f \to +\infty$ as $r \to 0$, so $r = 0$ is *timelike* (grey dashed) and the diagram is Kerr-like, repeating infinitely in both time directions; r_- is a Cauchy horizon for region I. **(c)** Extremal: here $f = (r^2 - r_+^2)^2/\ell^2 r^2 \geq 0$ *everywhere*, so both $r > r_+$ and $0 < r < r_+$ are static; the degenerate horizon lies at infinite tortoise *and* infinite proper radial distance [checked numerically], so there is no bifurcation surface — the two null edges of region I meet only at an internal infinity (open circle), the "infinite throat". We draw only this local gluing of one exterior to one interior region; for the full infinite chain see [15], Fig. 1c. Nothing in any panel is a curvature singularity: $R = -6/\ell^2$ throughout.

Normalisation check. Take static BTZ ($J = 0$, so $\pi^{ij} = 0$), $h_{ij}\mathrm{d}x^i\mathrm{d}x^j = \mathrm{d}r^2/f + r^2\mathrm{d}\varphi^2$ with $f = r^2/\ell^2 - 8GM$, lapse $\zeta^\perp = \sqrt{f}$ and zero shift, and vary with respect to M. Then [checked symbolically]

$$\sqrt{\sigma}\, n_i\, G^{ijkl}\big(\zeta^\perp \nabla_j \delta h_{kl} - \partial_j \zeta^\perp \delta h_{kl}\big) = 8G\,\delta M \quad \textit{exactly, for every } r, \tag{13}$$

so the integral over φ gives $16\pi G\,\delta M$ and (11) returns $\delta H = \delta M$. Note that the integrand is r-independent, not merely convergent: this is the statement that the Brown–Henneaux fall-off of §3 makes the charge finite without any subtraction. Integrating (11) for $\zeta = \partial_t$ and $\zeta = \partial_\varphi$ with the Brown–Henneaux fall-off of §3 gives exactly M and J of (6), the angular momentum coming from the $2\zeta_k \delta\pi^{ik}$ term, with the additive normalisation fixed by declaring the $M = J = 0$ geometry to have zero mass [11, 12]. Note the immediate consequence: global AdS_3, obtained by

setting $\alpha = 1$ in (23) (lecture 1), has

$$M_{\mathrm{AdS}_3} = -\frac{1}{8G}, \tag{14}$$

so the vacuum sits at negative mass, separated from the black-hole branch by the whole family of conical defects. This gap is the gravitational avatar of the CFT_2 Casimir energy $-c/12$ on the cylinder.

1.4 Temperature

Let $\chi = \partial_t + \Omega_H \partial_\varphi$ be the horizon generator, null on $r = r_+$, with angular velocity

$$\Omega_H = -N^\varphi(r_+) = \frac{4GJ}{r_+^2} = \frac{r_-}{\ell\, r_+}. \tag{15}$$

Surface gravity from the standard formula $\kappa = \frac{1}{2} f'(r)\big|_{r_+}$ in the corotating frame:

$$\kappa = \frac{r_+^2 - r_-^2}{\ell^2 r_+}, \qquad \Longrightarrow \qquad \boxed{T = \frac{\kappa}{2\pi} = \frac{r_+^2 - r_-^2}{2\pi \ell^2 r_+}} \tag{16}$$

[checked symbolically] Equivalently, the Euclidean section $t \to -i t_E$ is smooth iff t_E has period $\beta = 2\pi\ell^2 r_+/(r_+^2 - r_-^2)$ (with the simultaneous identification $\varphi \to \varphi + i\beta\Omega_H$).

It is extremely useful to package this chirally. Define

$$T_L = \frac{r_+ - r_-}{2\pi\ell^2}, \qquad T_R = \frac{r_+ + r_-}{2\pi\ell^2}, \qquad \frac{1}{T} = \frac{1}{2}\Big(\frac{1}{T_L} + \frac{1}{T_R}\Big). \tag{17}$$

Extremality $r_+ = r_-$ is $T_L = 0$: a chiral, zero-temperature-in-one-sector state.

1.5 Entropy and the first law

The Bekenstein–Hawking entropy uses the horizon "area" — here the circumference $A = 2\pi r_+$:

$$\boxed{S = \frac{A}{4G} = \frac{2\pi r_+}{4G} = \frac{\pi r_+}{2G}} \tag{18}$$

The first law is then an identity. With $M = (r_+^2 + r_-^2)/8G\ell^2$, $J = r_+ r_-/4G\ell$, $S = \pi r_+/2G$, T as in (16) and $\Omega_H = r_-/\ell r_+$, one checks [checked symbolically, both independent variations]

$$\mathrm{d}M = T\,\mathrm{d}S + \Omega_H\,\mathrm{d}J. \tag{19}$$

Explicitly, $T\,\mathrm{d}S = \dfrac{(r_+^2 - r_-^2)}{4G\ell^2 r_+}\mathrm{d}r_+$ and $\Omega_H \mathrm{d}J = \dfrac{r_-^2 \mathrm{d}r_+/r_+ + r_-\mathrm{d}r_-}{4G\ell^2}$, whose sum is $\dfrac{r_+\mathrm{d}r_+ + r_-\mathrm{d}r_-}{4G\ell^2} = \mathrm{d}M$.

1.6 Free energy and the Hawking–Page transition

The Gibbs free energy in the grand canonical ensemble is

$$F = M - TS - \Omega_H J = -\frac{r_+^2 - r_-^2}{8G\ell^2} \qquad \text{[checked]}, \tag{20}$$

The canonical ensemble is a function of (T,Ω), and two saddles are compared at the *same* temperature, so the useful form eliminates $r_\pm$ using $\Omega_H = r_-/\ell r_+$ and $T = (r_+^2 - r_-^2)/2\pi\ell^2 r_+$, i.e. $r_+ = 2\pi\ell^2 T/(1-\Omega^2\ell^2)$:

$$\boxed{F_{\rm BTZ}(T,\Omega) = -\frac{\pi^2\ell^2\,T^2}{2G\big(1-\Omega^2\ell^2\big)} \;=\; -\frac{\pi^2\ell^2}{2G}\,T_L T_R \;=\; -\frac{\pi^2 c\,\ell}{3}\,T_L T_R} \tag{21}$$

[checked symbolically], where $T_{L,R} = (r_+ \mp r_-)/2\pi\ell^2$ as in §3.4 and $c = 3\ell/2G$. The chiral form is the memorable one. At $\Omega = 0$ this is $F_{\rm BTZ} = -\pi^2 c\,\ell T^2/3$, which is exactly the Stefan–Boltzmann free energy of a CFT_2, $F = -\frac{\pi c}{6}T^2 L$, on a circle of circumference $L = 2\pi\ell$ [checked].

The competing saddle at the same (T,Ω) is *thermal* AdS_3: global AdS_3 with Euclidean time identified. It has no horizon, so $S = 0$ and $J = 0$, and therefore

$$F_{\mathrm{AdS}_3}(T,\Omega) = M_{\mathrm{AdS}_3} = -\frac{1}{8G} = -\frac{c}{12\ell}, \tag{22}$$

the Casimir energy of the CFT on the circle. *This is independent of* T — at leading order in the classical (large-c) limit, which is all we are working to here; temperature dependence enters thermal AdS_3 only through the one-loop boundary-graviton determinant of §5. The dominant phase at a given temperature is the one with the lower free energy, so (21) and (22) are to be compared at fixed T.

Setting $F_{\rm BTZ}(T,0) = F_{\mathrm{AdS}_3}$ gives, in the convenient variable $x \equiv 2\pi\ell T$,

$$8G\,F_{\rm BTZ} = -x^2, \qquad 8G\,F_{\mathrm{AdS}_3} = -1, \qquad \Longrightarrow \qquad \boxed{x = 1: \quad T_{\rm HP} = \frac{1}{2\pi\ell}, \quad \beta_{\rm HP} = 2\pi\ell, \quad r_+ = \ell, \quad 8GM = 1.} \tag{23}$$

[checked] Hence (Fig. 2):

- $T < 1/2\pi\ell$**:** $-x^2 > -1$, so $F_{\mathrm{AdS}_3} < F_{\rm BTZ}$ and **thermal** AdS_3 **dominates**. The black hole exists as a solution but is thermodynamically disfavoured.
- $T > 1/2\pi\ell$**:** $F_{\rm BTZ} < F_{\mathrm{AdS}_3}$ and **the black hole dominates**, with $F \to -\infty$ as $T \to \infty$.

This is the **Hawking–Page transition** [29] in its AdS_3 form [18]. Two remarks worth carrying forward: the transition temperature $\beta = 2\pi\ell$ is the self-dual point of the modular transformation $\tau \to -1/\tau$ that exchanges the two solid-torus fillings, which is why thermal AdS_3 and BTZ are exchanged there; and the low-temperature phase has F independent of T, hence $S = -\partial F/\partial T = 0$ — a gapped, $\mathcal{O}(c^0)$ density of states, versus the $\mathcal{O}(c)$ Cardy growth above the transition. With rotation the same computation gives the transition locus $r_+^2 - r_-^2 = \ell^2$, i.e. $(2\pi\ell T_L)(2\pi\ell T_R) = 1$ [checked].

1.7 Cardy and the microscopic count

Anticipating §3, where we derive $c = 3\ell/2G$, define

$$L_0 = \frac{M\ell + J}{2}, \qquad \bar{L}_0 = \frac{M\ell - J}{2}, \tag{24}$$

so that $M\ell = L_0 + \bar{L}_0$ and $J = L_0 - \bar{L}_0$. Then

$$L_0 = \frac{(r_+ + r_-)^2}{16G\ell}, \qquad \bar{L}_0 = \frac{(r_+ - r_-)^2}{16G\ell}, \tag{25}$$

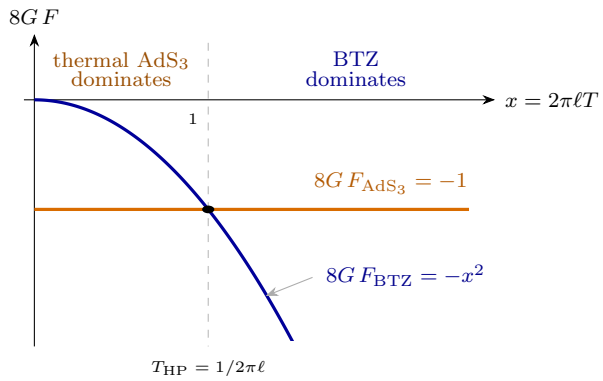


Figure 2: Free energies of the two saddles at the same temperature, in units $8G$ and with $x = 2\pi\ell T$ (non-rotating case). Thermal AdS_3 (orange) has $F = -1/8G$, independent of T at leading order in $1/c$; BTZ (blue) has $F = -\pi^2\ell^2T^2/2G$. They cross at $x = 1$, i.e. $\beta = 2\pi\ell$, $r_+ = \ell$: below it thermal AdS_3 has the lower free energy and dominates, above it the black hole does.

and the Cardy formula [28] gives

$$S_{\text{Cardy}} = 2\pi\sqrt{\frac{c\,L_0}{6}} + 2\pi\sqrt{\frac{c\,\bar{L}_0}{6}} = 2\pi\,\frac{r_+ + r_-}{8G} + 2\pi\,\frac{r_+ - r_-}{8G} = \frac{\pi r_+}{2G} = \frac{A}{4G}. \tag{26}$$

[checked symbolically] This is Strominger's derivation [17]: the Bekenstein–Hawking entropy of *any* BTZ black hole is reproduced by Cardy's asymptotic density of states in a CFT_2 with $c = \bar{c} = 3\ell/2G$, using only Brown–Henneaux and no string theory.

What the Cardy derivation does and does not establish (after Carlip [15, 30]).

(i) *There is no CFT here, only an algebra.* Brown–Henneaux gives the asymptotic symmetry *algebra* and its central charge. Cardy's formula additionally requires a unitary, modular-invariant CFT with a discrete spectrum. Strominger's step is to *assume* that such a theory exists and that BTZ states sit in it.

(ii) *The effective central charge matters.* Cardy's asymptotic growth is really $S = 2\pi\sqrt{c_{\text{eff}}\,L_0/6}$ with $c_{\text{eff}} = c - 24\Delta_{\min}$, where $\Delta_{\min}$ is the lowest weight in the spectrum. Getting $A/4G$ requires $\Delta_{\min} = -c/24$, i.e. that the *ground state be global* AdS_3 rather than the $M = 0$ geometry. This is a physical input, not an output.

(iii) *Liouville is a cautionary tale.* The asymptotic dynamics of AdS_3 gravity really is Liouville theory (§5), and it really has $c = 3\ell/2G$. But Liouville has a *continuous* spectrum bounded below by $\Delta \geq (c-1)/24$, so $c_{\text{eff}} = 1$: applying Cardy to Liouville gives $S \sim 2\pi\sqrt{L_0/6}$, nowhere near $A/4G$. A central charge alone does not determine a density of states.

(iv) *Where do the states live?* The counted states are boundary/asymptotic excitations at $r = \infty$, not obviously localised at the horizon. The derivation is insensitive to the interior, which is simultaneously its strength (universality — it works for any theory with these asymptotics, matter or no matter) and its weakness (it tells us nothing about what a black hole microstate *is*).

(v) *Universality is the real lesson.* The same $A/4G$ follows from several inequivalent arguments (near-horizon conformal symmetry, the Euclidean path integral, string microstate counting in D1–D5). That robustness suggests the answer is fixed by symmetry, which is encouraging, but it also means the agreement is weak evidence for any *particular* microscopic proposal.

Exercise 2. Cardy's formula is valid for $L_0 \gg c$. Show that this is the same as $r_+ \gg \ell$, i.e. large black holes. Show also that applying Cardy naively down to $L_0 = 0$ still gives the exact answer $A/4G$, and discuss why (hint: modular invariance and the vacuum at $L_0 = -c/24$).

2 The Fefferman–Graham expansion in three dimensions

2.1 Generalities and the $3d$ miracle

Any asymptotically locally AdS_{d+1} metric can be brought, in a neighbourhood of the conformal boundary, to Fefferman–Graham (FG) gauge [23, 24]

$$\mathrm{d}s^2 = \ell^2\,\frac{\mathrm{d}z^2}{z^2} + \frac{\ell^2}{z^2}\,g_{ij}(z,x)\,\mathrm{d}x^i\mathrm{d}x^j, \qquad g_{ij}(z,x) = g^{(0)}_{ij} + z^2 g^{(2)}_{ij} + z^4 g^{(4)}_{ij} + \dots \tag{27}$$

which is a genuine *gauge*: $g_{zz} = \ell^2/z^2$ and $g_{zi} = 0$ can be reached by a bulk diffeomorphism. Einstein's equations then determine $g^{(2)}, g^{(4)}, \dots$ algebraically/differentially in terms of $g^{(0)}$, except for one piece of free data at order z^d — the boundary stress tensor.

In general d the series does not terminate, and for even d it contains logarithms ($z^d \log z$) whose coefficient is the conformal anomaly [22]. In $d = 2$, i.e. $3d$ bulk, something special happens:

$$\boxed{g_{ij}(z,x) = g^{(0)}_{ij} + z^2\,g^{(2)}_{ij} + z^4\,\tfrac{1}{4}\left(g^{(2)}(g^{(0)})^{-1}g^{(2)}\right)_{ij} \quad \textit{exactly} \text{ — the expansion terminates.}} \tag{28}$$

The remaining content of the equations of motion is most transparent once $g^{(2)}$ is traded for the **boundary stress tensor**. Write $\operatorname{Tr} g^{(2)} \equiv (g^{(0)})^{ij} g^{(2)}_{ij}$ and define [20, 22]

$$\boxed{T_{ij} \;\equiv\; \frac{\ell}{8\pi G}\Big(g^{(2)}_{ij} - g^{(0)}_{ij}\operatorname{Tr} g^{(2)}\Big).} \tag{29}$$

Einstein's equations then reduce to exactly the two Ward identities of the dual CFT_2,

$$\boxed{\nabla^i_{(0)}\,T_{ij} = 0, \qquad T^{\,i}{}_{i} = \frac{c}{24\pi}\,R\big[g^{(0)}\big], \qquad c = \frac{3\ell}{2G}.} \tag{30}$$

[checked symbolically] The first is invariance under boundary diffeomorphisms; the second is the trace (conformal) anomaly, with precisely the Brown–Henneaux central charge that we will derive independently in §3. In terms of $g^{(2)}$ the same two statements read

$$\operatorname{Tr} g^{(2)} = -\tfrac{1}{2}\,R\big[g^{(0)}\big], \qquad \nabla^i_{(0)} g^{(2)}_{ij} = \partial_j \operatorname{Tr} g^{(2)}, \tag{31}$$

and (29) is the change of variables that turns them into (30). Note the structure: $g^{(0)}$ is the boundary metric — non-dynamical data, the "source" — while $g^{(2)}$ is the free data, the "response". The absence of higher terms in (27) says that in $3d$ there is no bulk graviton to generate an infinite tower of normalisable data beyond this one response; the "holographic RG flow" is trivial.

For the flat boundary metric used from §2.2 onwards, $\operatorname{Tr} g^{(2)} = 0$ and (29) collapses to $T_{\pm\pm} = (\ell/8\pi G)\,g^{(2)}_{\pm\pm}$, which is the normalisation that will reappear in the canonical charge (48).

Scope of the verification. The termination of (27) is checked here explicitly for flat $g^{(0)}$, in §2.2, where the resulting metric solves $R_{\mu\nu} = -\frac{2}{\ell^2} g_{\mu\nu}$ for arbitrary $\mathcal{L}(x^+), \bar{\mathcal{L}}(x^-)$. For general $g^{(0)}$ it is quoted from [21, 22, 19]. Every $2d$ metric is conformally flat, so the general case follows from $g^{(0)} = -e^{2\omega(x^+,x^-)} \mathrm{d}x^+ \mathrm{d}x^-$ with

$$g^{(2)}_{++} = \mathcal{L} - (\partial_+\omega)^2 + \partial_+^2\omega, \quad g^{(2)}_{--} = \bar{\mathcal{L}} - (\partial_-\omega)^2 + \partial_-^2\omega, \quad g^{(2)}_{+-} = \partial_+\partial_-\omega, \tag{32}$$

for which (30) holds [checked symbolically]; the Schwarzian-like inhomogeneous terms are the familiar anomalous transformation of a CFT_2 stress tensor under a Weyl rescaling of the boundary metric.

2.2 Bañados metrics

Take the flat boundary metric $g^{(0)} = -\mathrm{d}x^+\mathrm{d}x^-$ on the cylinder. Then (30) force $g^{(2)}$ to be traceless and chirally conserved, so $g^{(2)}_{++} = \mathcal{L}(x^+)$, $g^{(2)}_{--} = \bar{\mathcal{L}}(x^-)$, $g^{(2)}_{+-} = 0$, with $\mathcal{L}, \bar{\mathcal{L}}$ *arbitrary* functions of one variable. The resulting exact solution is the Bañados family [19]:

$$\boxed{\mathrm{d}s^2 = \ell^2 \left[\frac{\mathrm{d}\rho^2}{\rho^2} + \mathcal{L}(x^+)\,(\mathrm{d}x^+)^2 + \bar{\mathcal{L}}(x^-)\,(\mathrm{d}x^-)^2 - \Big(\frac{1}{\rho^2} + \rho^2 \mathcal{L}\bar{\mathcal{L}}\Big)\mathrm{d}x^+\mathrm{d}x^- \right]} \tag{33}$$

with $\rho \to 0$ the conformal boundary. [checked symbolically: $R_{\mu\nu} = -\frac{2}{\ell^2} g_{\mu\nu}$ and $R = -6/\ell^2$ for arbitrary $\mathcal{L}(x^+), \bar{\mathcal{L}}(x^-)$.]

Equation (33) is the *most general* solution of $3d$ vacuum Einstein gravity with a flat boundary metric and Brown–Henneaux boundary conditions. The entire phase space of the theory is the space of pairs of periodic functions $(\mathcal{L}, \bar{\mathcal{L}})$, modulo the identifications generated by the asymptotic symmetries of §3. By (29) these are the boundary stress tensor components, $T_{\pm\pm} = (\ell/8\pi G)\{\mathcal{L}, \bar{\mathcal{L}}\}$: in CFT language $\mathcal{L}$ and $\bar{\mathcal{L}}$ are the holomorphic and antiholomorphic stress tensors, and the phase space is a pair of Virasoro coadjoint orbits — a structure we return to in §5.

From FG gauge to the standard gauges. The map to the coordinates of §1 is a redefinition of the radial variable alone. Define

$$\boxed{\frac{r^2}{\ell^2} = \frac{\big(1+\rho^2\mathcal{L}\big)\big(1+\rho^2\bar{\mathcal{L}}\big)}{\rho^2} = \frac{1}{\rho^2} + \mathcal{L} + \bar{\mathcal{L}} + \rho^2\mathcal{L}\bar{\mathcal{L}}} \tag{34}$$

(for constant $\mathcal{L}, \bar{\mathcal{L}}$; when $\mathcal{L}$ depends on x^+ the change of coordinates also shifts $x^\pm$). Then (33) becomes (4) with t, φ unchanged, and [checked symbolically: $g_{rr} f = 1$ identically, and $g_{t\varphi}/g_{\varphi\varphi} = N^\varphi$]

$$\frac{r_\pm}{\ell} = \sqrt{\mathcal{L}} \pm \sqrt{\bar{\mathcal{L}}}, \qquad \Longleftrightarrow \qquad \mathcal{L} = \frac{(r_+ + r_-)^2}{4\ell^2}, \quad \bar{\mathcal{L}} = \frac{(r_+ - r_-)^2}{4\ell^2}. \tag{35}$$

Combining with (6) gives the dictionary used throughout the rest of the course:

$$\boxed{\mathcal{L} = 2G\Big(M + \frac{J}{\ell}\Big), \qquad \bar{\mathcal{L}} = 2G\Big(M - \frac{J}{\ell}\Big), \qquad L_0 = \frac{c}{6}\,\mathcal{L}, \quad \bar{L}_0 = \frac{c}{6}\,\bar{\mathcal{L}}, \quad c = \frac{3\ell}{2G}.} \tag{36}$$

[checked symbolically] Since (34) holds for any constant $(\mathcal{L}, \bar{\mathcal{L}})$, one map covers every stationary solution at once: global AdS_3, the conical defects and the BTZ black holes are recovered by inserting the corresponding values of $\mathcal{L}, \bar{\mathcal{L}}$, listed in §2.3.

Exercise 3. Carry out the substitution (34) for constant $\mathcal{L}, \bar{\mathcal{L}}$ and verify $g_{rr} = 1/f(r)$ with f as in (5). (Hint: show first that $\ell^2 r^2 f = \ell^4\big(\mathcal{L}\bar{\mathcal{L}}\rho^2 - \rho^{-2}\big)^2$.) Then check that $\mathcal{L} = \bar{\mathcal{L}} = -\frac{1}{4}$ returns global AdS_3 in the form (15) (lecture 1).

2.3 The phase space and its supersymmetric loci

Set the dimensionless coordinates

$$x \equiv 8GM = 2(\mathcal{L} + \bar{\mathcal{L}}), \qquad y \equiv \frac{8GJ}{\ell} = 2(\mathcal{L} - \bar{\mathcal{L}}), \tag{37}$$

so that $\mathcal{L} = (x+y)/4$ and $\bar{\mathcal{L}} = (x-y)/4$. The constant-$(\mathcal{L}, \bar{\mathcal{L}})$ sector of the phase space is then the (x, y) plane (Fig. 3):

- $\mathcal{L}, \bar{\mathcal{L}} \geq 0$ (i.e. $x \geq |y|$): **BTZ black holes**, with the boundary rays $y = \pm x$ the **extremal** black holes $|J| = M\ell$.
- $-\frac{1}{4} \leq \mathcal{L}, \bar{\mathcal{L}} < 0$: **conical defects** (spinning point particles), the square with vertices $(0,0)$, $(-\frac{1}{2}, \pm\frac{1}{2})$, $(-1, 0)$.
- $\mathcal{L} = \bar{\mathcal{L}} = -\frac{1}{4}$, i.e. $(x, y) = (-1, 0)$: **global** AdS_3, the ground state.
- $\mathcal{L} = \bar{\mathcal{L}} = 0$, i.e. the origin: the **massless BTZ** geometry, the $\mathcal{L} \to 0$ limit of black holes and the $\alpha \to 0$ limit of defects.
- $\mathcal{L}\bar{\mathcal{L}} < 0$: **overspinning**, $|J| > M\ell$. Excluded: no horizon and a naked CTC region (Remark 1(b)). $\mathcal{L} < -\frac{1}{4}$ or $\bar{\mathcal{L}} < -\frac{1}{4}$: conical *excess*, also excluded. Note that the retained spinning defects are not free of CTCs either — see Remark 1(c); what excludes the overspinning wedge is that it is neither a black hole nor a point particle.

Supersymmetry. Once the theory is embedded in $\mathcal{N}$-extended AdS_3 supergravity (matter included), one can ask which of these backgrounds admit Killing spinors,

$$\Big(\nabla_\mu + \tfrac{1}{2\ell}\gamma_\mu\Big)\epsilon = 0. \tag{38}$$

It is essential to distinguish two questions, because (38) is a *local* equation while supersymmetry of a background is a *global* statement:

(a) Does (38) have solutions locally? Since every solution here is locally AdS_3, the answer is always yes, and there are always four of them. This is no information at all.

(b) Do any of those solutions descend to the quotient $\widetilde{AdS_3}/\Gamma$ as *single-valued* spinor fields? This depends on (i) the holonomy of Γ acting on spinors and (ii) the *choice of spin structure*, and it is here that the various solutions differ.

Coussaert and Henneaux [13] answered (b) for the vacuum BTZ family:

- **Global** AdS_3 ($x = -1$): **2** Killing spinors, *antiperiodic* in φ (NS sector). This is the maximally supersymmetric vacuum.
- **Massless BTZ** $M = J = 0$ (origin): **2** Killing spinors, *periodic* in φ (R sector).

- **Extremal BTZ** $J = \pm M\ell$, $M > 0$: **1** Killing spinor, also periodic. These are the BPS states of the boundary theory.
- Generic non-extremal BTZ and generic conical defects: **no** Killing spinors.

Is the $M = J = 0$ supersymmetry really global? Yes — but only with a spin structure that has to be chosen, and the choice is not forced. The point is which spin structures the geometry admits.

- In **global** AdS_3 the circle $\varphi \simeq \varphi + 2\pi$ is *contractible* (it shrinks smoothly at the origin $r = 0$). A contractible circle admits only one spin structure, the antiperiodic (NS) one. The AdS Killing spinors happen to be antiperiodic, so they survive: the supersymmetry is genuinely global, with no choice involved.
- In the $M = J = 0$ **geometry** the identification is by a *parabolic* element and $r = 0$ is a null surface, not a regular origin: the φ-circle is *not* contractible. Both spin structures therefore exist, and Coussaert–Henneaux's two Killing spinors are periodic. So the statement is precise only as: the background ($M = J = 0$ *geometry, Ramond spin structure*) preserves two supercharges. Equip the same metric with the NS spin structure and *all* supersymmetry is broken — the would-be Killing spinors are then double-valued. Nothing is inconsistent here; the spin structure is independent data in the path integral, and both choices are legitimate backgrounds. (This is precisely the bulk statement of the CFT fact that the $M = 0$ geometry is the R-sector ground state while global AdS_3 is the NS-sector one, related by spectral flow.)
- The same caveat attaches to the **extremal** BTZ Killing spinor: it is periodic, so it exists on the R spin structure only.
- For **conical defects** with $\alpha \neq 1$ the circle *is* contractible, so the spinor must be antiperiodic under $\varphi \to \varphi + 2\pi$; but the local AdS Killing spinor returns with a phase $e^{\pm i\pi\alpha}$, which is single-valued only for special α. In pure $\mathcal{N} = (1, 1)$ supergravity this kills supersymmetry for generic α. It can be restored if there is an extra R-symmetry connection whose holonomy cancels the phase — a graviphoton Wilson line in $(2, 0)$ supergravity [14], or a compensating rotation on the S^3 in the D1–D5 uplift, which is why the $\alpha = 1/n$ orbifolds are supersymmetric there [25].

[The spin-structure statements above follow from the periodicity properties reported in [13] together with the standard contractibility argument; we have not re-derived the Killing spinors themselves.]

Izquierdo and Townsend [14] extended the analysis to $(2, 0)$ AdS_3 supergravity with matter. There the conical branch is no longer entirely non-supersymmetric: switching on the graviphoton Wilson line, families of supersymmetric conical spaces appear — including geometries of arbitrarily large negative energy with only conical singularities — as well as the fractionally charged point-particle spacetimes. In the D1–D5 realisation of AdS_3 these supersymmetric conical defects are the $\mathbb{Z}_n$ orbifolds $\alpha = 1/n$, which are the classical geometries dual to particular R-sector ground states [25]. We indicate the corresponding loci in Fig. 3 with the caveat that the precise pattern depends on the amount of supergravity and on the matter content; the four bullets above are the model-independent statements.

Exercise 4. Show that the mass gap between global AdS_3 and the lightest black hole is $\Delta M = 1/8G$, and that in CFT variables this is $\Delta(L_0 + \bar{L}_0) = c/12$. This "BTZ threshold" is a statement about the *spectrum*, distinct from the Hawking–Page transition of §1.6, which concerns which saddle dominates the canonical ensemble at a given T: the threshold sits at $M = 0$, the transition at $8GM = 1$, i.e. $r_+ = \ell$.

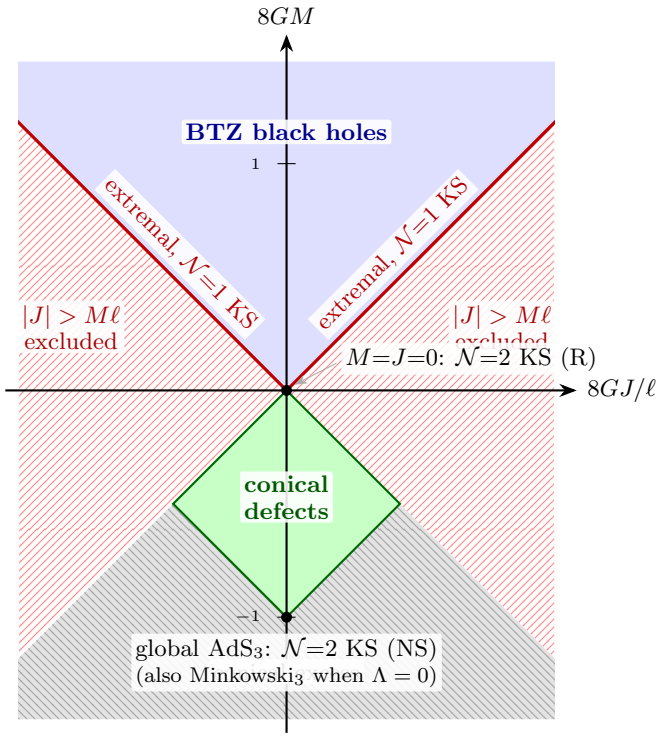


Figure 3: Phase space of constant-$(\mathcal{L}, \bar{\mathcal{L}})$ solutions, with the mass on the vertical axis: $\mathcal{L} = \big(8GM + 8GJ/\ell\big)/4$, $\bar{\mathcal{L}} = \big(8GM - 8GJ/\ell\big)/4$. Blue: BTZ black holes, $\mathcal{L}, \bar{\mathcal{L}} \geq 0$. Green: conical defects, $-\frac{1}{4} \leq \mathcal{L}, \bar{\mathcal{L}} < 0$. The bottom vertex $8GM = -1$ is global AdS_3 (2 Killing spinors, antiperiodic); the origin is massless BTZ (2 Killing spinors, periodic); the red rays $|J| = M\ell$ are extremal BTZ (1 Killing spinor) [13]. Hatched regions are excluded: naked CTCs ($|J| > M\ell$) and conical excess. With matter, additional supersymmetric loci appear inside the defect region [14, 25]. As shown in §2.4 (lecture 1), the $\Lambda = 0$ theory occupies exactly the segment $-1 \leq 8GM \leq 0$, $J = 0$ of this diagram, with Minkowski$_3$ at the same point as global AdS_3.

3 Asymptotic symmetries and the Brown–Henneaux central charge

3.1 What an asymptotic symmetry group *is*

Section 1.1 (lecture 1) established that $3d$ gravity has no local degrees of freedom: the Weyl tensor vanishes, every solution is locally AdS_3, and there is no graviton to propagate. Taken at face value this would leave a theory with nothing in it. It does not, and the reason is the subject of this section.

A spacetime with a boundary admits diffeomorphisms that do not die off at that boundary. Some of these carry non-vanishing canonical charge, and therefore relate physically inequivalent configurations rather than acting as redundancies. The group they generate — the **asymptotic symmetry group** — is then a genuine global symmetry of the theory, and acting with it on a given solution sweeps out a family of physically distinct solutions. These are the **boundary gravitons**. They are the entire dynamical content of $3d$ gravity: the bulk carries nothing, the boundary carries everything. The Bañados phase space (33), parametrised by two arbitrary functions $\mathcal{L}(x^+), \bar{\mathcal{L}}(x^-)$, is precisely the orbit of global AdS_3 and of the BTZ black holes under this group, and We follow the logic set out in the Kerr/CFT paper of Guica, Hartman, Song and

Strominger [1] and, in its general covariant-phase-space form, Barnich–Brandt [2].

In a gauge theory the gauge group is not a symmetry: it is a redundancy. But in a theory on a manifold *with a boundary*, not all gauge transformations are redundancies. Once one fixes boundary conditions, the diffeomorphisms split into three classes:

(i) **Disallowed** — those that violate the boundary conditions. These are simply not in the theory.

(ii) **Allowed but trivial** — those preserving the boundary conditions whose associated canonical charge *vanishes*. These act trivially on the physical phase space: they are proper gauge transformations, pure redundancy.

(iii) **Allowed and non-trivial** — those preserving the boundary conditions with *non-vanishing* charge. These map physically distinct states into one another. They are global symmetries.

The **asymptotic symmetry group** is the quotient

$$\boxed{\text{ASG} = \frac{\{\text{diffeomorphisms preserving the boundary conditions}\}}{\{\text{diffeomorphisms with vanishing canonical charge}\}}} \tag{39}$$

Its Lie algebra is realised on phase space by the charges themselves, under the Dirac bracket, and this realisation may be *centrally extended* — which is the whole point. A classical central charge is not an anomaly; it is the statement that the charges do not vanish on the reference solution, exactly like the central term $\{x, p\} = 1$ in the Heisenberg algebra of translations.

Boundary conditions are a physical choice, and a delicate one. Nothing above determines the boundary conditions; they are input. A good choice must simultaneously satisfy four requirements, and they pull against each other:

(1) **Inclusive enough.** The solutions of physical interest — here global AdS_3, all conical defects and all BTZ black holes — must satisfy them. Too tight a condition and one is left with a theory containing nothing.

(2) **Finite charges.** The surface integrals defining $Q[\xi]$ must converge as $r \to \infty$. Too loose a condition and the energy of a generic allowed configuration diverges.

(3) **Integrable charges.** The canonical analysis yields $\delta Q[\xi]$, a one-form on phase space. For a charge $Q[\xi]$ to exist at all, this one-form must be *integrable*, $\delta_1(\delta_2 Q) - \delta_2(\delta_1 Q) = 0$. Sloppy boundary conditions produce non-integrable δQ, and then there is no conserved charge — only a flux. (This is the standard situation at null infinity in $4d$; it is what makes Bondi mass loss subtle.)

(4) **Non-trivial ASG.** If the boundary conditions are so restrictive that every allowed diffeomorphism has vanishing charge, the ASG is trivial and one has learned nothing. Conversely, if they are so weak that the ASG is all of Diff, the "symmetry" carries no information either.

The Brown–Henneaux conditions of §3.2 satisfy all four, and the resulting algebra is infinite-dimensional.

Remark 3. The choice is not unique: other consistent boundary conditions on AdS_3 yield different asymptotic symmetry algebras, for instance $\mathfrak{u}(1)$ current algebras or the warped-conformal algebra of warped AdS_3. Brown–Henneaux is the choice for which the dual is an ordinary CFT_2 on a fixed flat cylinder, and it is the one we use throughout.

3.2 The Brown–Henneaux boundary conditions

Brown and Henneaux [9] proposed the fall-off conditions (in the coordinates of (15) (lecture 1))

$$\begin{aligned} g_{tt} &= -\frac{r^2}{\ell^2} + \mathcal{O}(1), \quad g_{t\varphi} = \mathcal{O}(1), \qquad g_{tr} = \mathcal{O}(r^{-3}), \\ g_{\varphi\varphi} &= r^2 + \mathcal{O}(1), \qquad g_{\varphi r} = \mathcal{O}(r^{-3}), \quad g_{rr} = \frac{\ell^2}{r^2} + \mathcal{O}(r^{-4}). \end{aligned} \tag{40}$$

Let us check them against the four criteria of §3.1.

(1) *Inclusive.* The leading terms are those of global AdS_3; the $\mathcal{O}(1)$ pieces of $g_{tt}, g_{t\varphi}, g_{\varphi\varphi}$ are exactly where M and J live, so the whole two-parameter BTZ family and all conical defects are included. Equivalently: **in FG language (40) says that the solution is of Bañados form (33), with the boundary metric $g^{(0)}$ fixed to be the flat cylinder and only $\mathcal{L}, \bar{\mathcal{L}}$ free.** Since (33) is the general solution, nothing is lost.

(2) *Finite.* Each relaxation is exactly one power of r^2 below the leading behaviour — the "normalisable" mode. Weakening $g_{\varphi\varphi}$ to $r^2 + \mathcal{O}(r)$, say, would make the mass integral diverge.

(3) *Integrable.* Because $g^{(0)}$ is held *fixed* (non-dynamical), the variation δQ involves only $\delta\mathcal{L}$ and $\delta\bar{\mathcal{L}}$, and (48) below is manifestly a total variation. Had we allowed $g^{(0)}$ to fluctuate — i.e. made the boundary metric dynamical — δQ would acquire a $T^{ij}\delta g^{(0)}_{ij}$ flux term and cease to be integrable. *This is the technical reason the boundary metric must be frozen.*

(4) *Non-trivial.* We are about to find that the ASG is two copies of the Virasoro algebra, with a central charge. Note that the exact isometry group $SO(2,2)$ sits inside it as the $\{L_0, L_{\pm1}\} \times \{\bar{L}_0, \bar{L}_{\pm1}\}$ subalgebra: the ASG is infinite-dimensional while the isometry group is six-dimensional.

3.3 Asymptotic Killing vectors and the Witt algebra

We look for the vector fields preserving (33). Preserving $g_{\rho\rho} = \ell^2/\rho^2$ and $g_{\rho\pm} = 0$ determines ξ completely in terms of two arbitrary functions $\varepsilon(x^+), \bar{\varepsilon}(x^-)$:

$$\boxed{\begin{aligned} \xi^\rho &= \frac{\rho}{2}\big(\varepsilon' + \bar{\varepsilon}'\big), \\ \xi^+ &= \varepsilon + \frac{\rho^2\,\bar{\varepsilon}'' + \rho^4\,\bar{\mathcal{L}}\,\varepsilon''}{2\big(1 - \rho^4\mathcal{L}\bar{\mathcal{L}}\big)}, \\ \xi^- &= \bar{\varepsilon} + \frac{\rho^2\,\varepsilon'' + \rho^4\,\mathcal{L}\,\bar{\varepsilon}''}{2\big(1 - \rho^4\mathcal{L}\bar{\mathcal{L}}\big)}. \end{aligned}} \tag{41}$$

[checked symbolically: $\mathcal{L}_\xi g_{\rho\rho} = \mathcal{L}_\xi g_{\rho\pm} = 0$ exactly, for arbitrary $\mathcal{L}(x^+), \bar{\mathcal{L}}(x^-), \varepsilon(x^+), \bar{\varepsilon}(x^-)$.] This is *exact*, not asymptotic: (41) generates an honest symmetry of the whole Bañados phase space.

Equation (41) is where the boundary gravitons of §3.1 become concrete. Acting with ξ on a background produces the linearised perturbation $h_{\mu\nu} = \mathcal{L}_\xi\, \bar{g}_{\mu\nu}$, which solves the linearised Einstein equations by construction. Because ξ does not vanish at the boundary, this perturbation is not pure gauge: it carries finite, non-zero canonical charge (computed in §3.4), so $\bar{g} + h$ is a physically distinct solution from $\bar{g}$. Such configurations are the **boundary gravitons**, and by the argument of §1.1 (lecture 1) they exhaust the degrees of freedom of the theory — there is nothing else for

$3d$ gravity to excite. Finite transformations then move one through the Bañados family (33), so the space of $(\mathcal{L}, \bar{\mathcal{L}})$ is the space of boundary-graviton states built on a given background. Which background one starts from matters: as we will see in §5, the orbit of global AdS_3 and the orbit of a BTZ black hole are different coadjoint orbits, with different stabilisers. The action on the data is

$$\boxed{\delta_\varepsilon \mathcal{L} = \varepsilon\, \mathcal{L}' + 2\varepsilon'\, \mathcal{L} - \tfrac{1}{2}\, \varepsilon''', \qquad \delta_{\bar\varepsilon} \bar{\mathcal{L}} = \bar\varepsilon\, \bar{\mathcal{L}}' + 2\bar\varepsilon'\, \bar{\mathcal{L}} - \tfrac{1}{2}\, \bar\varepsilon'''.} \tag{42}$$

[checked symbolically] This is precisely the transformation of a CFT_2 stress tensor: a weight-2 primary plus an *anomalous* inhomogeneous Schwarzian-derivative piece. The ε''' term is the classical central extension, and the central charge computed in §3.4 follows from its coefficient.

Consistency check on the sign. Global AdS_3 ($\mathcal{L} = \bar{\mathcal{L}} = -\frac{1}{4}$) must be invariant under the six exact isometries, which correspond to $\varepsilon \in \{1, e^{\pm ix^+}\}$ and $\bar\varepsilon \in \{1, e^{\pm ix^-}\}$. Indeed, for $\varepsilon = e^{ix^+}$, (42) gives $2i(-\frac{1}{4}) + \frac{1}{2}i = 0$. With the opposite sign of the ε''' term one would instead find $\mathcal{L} = +\frac{1}{4}$ as the invariant point, which is not AdS. [checked]

The algebra of the vector fields: two copies of Witt

Before any charges, the vector fields (41) already close into an algebra, and it is worth isolating that step: *the algebra of the ASG is fixed by the vector fields alone; only its central extension requires the charges.*

There is one subtlety. The vector field (41) depends on the solution through $\mathcal{L}, \bar{\mathcal{L}}$ — it is *field-dependent*. For such generators the correct bracket is not the naive Lie bracket but the *modified* (adjusted) bracket of Barnich–Troessaert [3],

$$\big[\xi_1, \xi_2\big]_* \;\equiv\; \big[\xi_1, \xi_2\big]_{\text{Lie}} - \delta_{\xi_1}\xi_2 + \delta_{\xi_2}\xi_1, \tag{43}$$

where $\delta_{\xi_1}\xi_2$ means varying the explicit $\mathcal{L}, \bar{\mathcal{L}}$ appearing in ξ_2 according to (42). With this bracket one finds, for the exact vectors (41),

$$\boxed{\big[\xi(\varepsilon_1, \bar\varepsilon_1),\, \xi(\varepsilon_2, \bar\varepsilon_2)\big]_* = \xi\big(\varepsilon_{12},\, \bar\varepsilon_{12}\big), \qquad \varepsilon_{12} = \varepsilon_1\varepsilon_2' - \varepsilon_2\varepsilon_1', \quad \bar\varepsilon_{12} = \bar\varepsilon_1\bar\varepsilon_2' - \bar\varepsilon_2\bar\varepsilon_1'.} \tag{44}$$

[checked symbolically: all three components, for arbitrary $\mathcal{L}(x^+)$ and arbitrary $\varepsilon_1, \varepsilon_2$.] Note that ε_{12} involves only the unbarred functions and $\bar\varepsilon_{12}$ only the barred ones: **the two sectors commute**.

Now expand in modes. Set

$$\ell_m \equiv \xi\big(\varepsilon = e^{imx^+},\, \bar\varepsilon = 0\big), \qquad \bar\ell_m \equiv \xi\big(\varepsilon = 0,\, \bar\varepsilon = e^{imx^-}\big). \tag{45}$$

Then $\varepsilon_{12} = e^{imx^+}(in)e^{inx^+} - e^{inx^+}(im)e^{imx^+} = i(n-m)e^{i(m+n)x^+}$, so

$$[\ell_m, \ell_n]_* = i(n-m)\, \ell_{m+n}. \tag{46}$$

Absorbing the factor of i by defining $L_m \equiv i\,\ell_m$, $\bar L_m \equiv i\,\bar\ell_m$, we obtain

$$\boxed{[L_m, L_n] = (m-n)L_{m+n}, \qquad [\bar L_m, \bar L_n] = (m-n)\bar L_{m+n}, \qquad [L_m, \bar L_n] = 0.} \tag{47}$$

This is **two commuting copies of the Witt algebra** $\mathrm{Diff}(S^1)$ — the Virasoro algebra *without* a central extension. Two remarks:

- Restricting to $m \in \{-1, 0, +1\}$ gives $\mathfrak{sl}(2,\mathbb{R}) \oplus \mathfrak{sl}(2,\mathbb{R}) = \mathfrak{so}(2,2)$: the exact isometries of AdS_3 are the globally well-defined subalgebra of (47), exactly as the global conformal group sits inside Virasoro in CFT_2. The ASG is infinite-dimensional; the isometry group is six-dimensional.
- No central term can appear in (47): a central extension is not a property of vector fields, which act on the manifold, but of their *representation by charges* on phase space. That is the content of the next subsection.

Exercise 5. Verify (44) at leading order in ρ by hand (i.e. using $\xi^\rho = \frac{\rho}{2}\varepsilon'$, $\xi^+ = \varepsilon$, $\xi^- = \frac{\rho^2}{2}\varepsilon''$). Then convince yourself that the naive Lie bracket *fails* to close on (41) once the $\mathcal{L}$-dependent terms are kept, and that (43) repairs it.

3.4 The charges and the central charge

The Regge–Teitelboim charge generating (41) is (normalisation fixed by demanding $Q[\partial_t] = M$ using (36)):

$$Q[\varepsilon, \bar{\varepsilon}] = \frac{\ell}{8\pi G} \int_0^{2\pi} \mathrm{d}\varphi \, \big[\varepsilon \, \mathcal{L} + \bar{\varepsilon} \, \bar{\mathcal{L}}\big]. \tag{48}$$

Check: $\partial_t = \frac{1}{\ell}(\partial_+ + \partial_-)$ means $\varepsilon = \bar{\varepsilon} = 1/\ell$, giving $Q = \frac{\ell}{8\pi G} \cdot \frac{2\pi}{\ell}(\mathcal{L} + \bar{\mathcal{L}}) = \frac{\mathcal{L}+\bar{\mathcal{L}}}{4G} = M$ by (36). ✓

Define modes at $t = 0$ (where $x^\pm = \pm\varphi$; take the + sector)

$$L_m = \frac{\ell}{8\pi G} \int_0^{2\pi} \mathrm{d}\varphi \, e^{im\varphi} \, \mathcal{L}. \tag{49}$$

For global AdS_3, $\mathcal{L} = -\frac{1}{4}$ and $L_0 = \frac{\ell}{8\pi G}(2\pi)(-\frac{1}{4}) = -\frac{\ell}{16G}$. Now use (42) with $\varepsilon_1 = e^{im\varphi}$, $\varepsilon_2 = e^{in\varphi}$:

$$\delta_{\varepsilon_1} Q[\varepsilon_2] = \frac{\ell}{8\pi G} \int \mathrm{d}\varphi \, e^{in\varphi} \Big(e^{im\varphi} \mathcal{L}' + 2(im) e^{im\varphi} \mathcal{L} - \tfrac{1}{2}(im)^3 e^{im\varphi} \Big). \tag{50}$$

The first two terms integrate by parts to $i(m-n)L_{m+n}$. The last term gives

$$-\frac{\ell}{16\pi G}(im)^3 \int_0^{2\pi} \mathrm{d}\varphi \, e^{i(m+n)\varphi} = -\frac{\ell}{16\pi G}\,(-im^3)\, 2\pi\delta_{m+n,0} = \; i\,\frac{\ell\, m^3}{8G}\,\delta_{m+n,0}. \tag{51}$$

Stripping the overall i (Dirac bracket $\to$ commutator, $i\{\,,\,\} \to [\,,\,]$),

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{\ell}{8G}\, m^3\, \delta_{m+n,0} \;\equiv\; (m-n)L_{m+n} + \frac{c}{12} m^3 \delta_{m+n,0}, \tag{52}$$

whence

$$\boxed{c = \bar{c} = \frac{12\,\ell}{8G} = \frac{3\ell}{2G}} \tag{53}$$

[checked: the arithmetic above and the normalisation $Q[\partial_t] = M$.] Shifting $L_0 \to L_0 + \frac{c}{24}$ so that global AdS_3 has $L_0 = 0$ turns $m^3 \to m^3 - m$ in (52), the standard form, and confirms

$$L_0^{\mathrm{AdS}_3} = -\frac{c}{24} = -\frac{\ell}{16G}, \tag{54}$$

matching the direct computation above. ✓

This is the Brown–Henneaux result [9]: the asymptotic symmetry algebra of AdS_3 gravity is *two copies of the Virasoro algebra* with $c = 3\ell/2G$, not merely $\mathrm{SO}(2,2)$. Twelve years before AdS/CFT was formulated, this was already the statement that quantum gravity on AdS_3 is a two-dimensional CFT.

Exercise 6. Redo the derivation in the φ-independent constant-$\mathcal{L}$ sector and show that (48) reproduces both M and J of (6). Then show that $c = 3\ell/2G$ is equivalent to $S_{\text{Cardy}} = A/4G$, closing the loop with §1.

Exercise 7. Recompute c in the Chern–Simons formulation [8, 10], where the bulk action is $I = I_{\text{CS}}[A] - I_{\text{CS}}[\bar{A}]$ with $A = \omega + e/\ell$, $\bar{A} = \omega - e/\ell$ and CS level $k = \ell/4G$. Show $c = 6k$ and recover $3\ell/2G$. (We will develop this formulation properly in Week 3; it makes the two chiralities manifest and generalises to higher spin.)

3.5 Aside: the flat-space limit and BMS_3

It is natural to ask what happens to (52) as $\ell \to \infty$. Barnich and Compère [26] carried out the analogous analysis at *null* infinity of asymptotically flat $3d$ spacetimes. The asymptotic symmetry algebra is $\mathfrak{bms}_3$: the semi-direct sum of $\mathrm{Diff}(S^1)$ ("superrotations", generators L_n) with an abelian ideal of supertranslations M_n,

$$\begin{aligned} [L_m, L_n] &= (m-n)L_{m+n} + \frac{c_{LL}}{12} m^3 \delta_{m+n,0}, \\ [L_m, M_n] &= (m-n)M_{m+n} + \frac{c_{LM}}{12} m^3 \delta_{m+n,0}, \qquad c_{LL} = 0, \quad c_{LM} = \frac{3}{G}. \\ [M_m, M_n] &= 0, \end{aligned} \tag{55}$$

This is exactly the İnönü–Wigner contraction of the two Virasoros: setting $L_n^{\text{bms}} = L_n - \bar{L}_{-n}$ and $M_n = \frac{1}{\ell}(L_n + \bar{L}_{-n})$ and taking $\ell \to \infty$ with $c = 3\ell/2G$ fixed gives $c_{LM} = \lim 2c/\ell = 3/G$. The flat-space analogue of BTZ — the flat cosmological/shifted-boundary solutions — and a flat-space Cardy formula follow [26, 27].

Remark 4. Brown–Henneaux is an AdS_3 statement; there is no ℓ-dependent Virasoro central charge for asymptotically flat $3d$ gravity, only the $c_{LM} = 3/G$ above. We flag this because the two are often conflated.

4 The Chern–Simons formulation

Everything we have done so far was in metric language. There is a second formulation, due to Achúcarro–Townsend [8] and Witten [10], which makes the two chiralities manifest, trivialises the equations of motion, and is the natural starting point for higher-spin gravity, Wilson-line observables and defects. We introduce it here and will use it from Week 3 onward.

4.1 Gravity as two Chern–Simons theories

Work in first-order (Cartan) variables: a dreibein $e^a = e^a_\mu \mathrm{d}x^\mu$ and a dualised spin connection $\omega^a = \frac{1}{2}\varepsilon^{abc}\omega_{bc}$, $a = 0, 1, 2$, with $g_{\mu\nu} = \eta_{ab} e^a_\mu e^b_\nu$. The first-order action is

$$I = \frac{1}{8\pi G} \int \Big[e_a \wedge \big(\mathrm{d}\omega^a + \tfrac{1}{2}\varepsilon^{abc}\omega_b \wedge \omega_c \big) + \frac{1}{6\ell^2} \varepsilon^{abc} e_a \wedge e_b \wedge e_c \Big], \tag{56}$$

whose equations of motion are the torsion condition $T^a = \mathrm{d}e^a + \varepsilon^{abc}\omega_b \wedge e_c = 0$ and $R^a + \frac{1}{2\ell^2}\varepsilon^{abc} e_b \wedge e_c = 0$, i.e. constant curvature. Now define the two connections

$$\boxed{A^a = \omega^a + \frac{e^a}{\ell}, \qquad \bar{A}^a = \omega^a - \frac{e^a}{\ell}, \qquad A = A^a J_a, \quad \bar{A} = \bar{A}^a J_a, \quad J_a \in \mathfrak{sl}(2,\mathbb{R}).} \tag{57}$$

A short computation shows that the two equations of motion are equivalent to the vanishing of *both* curvatures,

$$F = \mathrm{d}A + A \wedge A = 0, \qquad \bar{F} = \mathrm{d}\bar{A} + \bar{A} \wedge \bar{A} = 0, \tag{58}$$

and that the action becomes a difference of two Chern–Simons actions,

$$\boxed{I = I_{\mathrm{CS}}[A] - I_{\mathrm{CS}}[\bar{A}], \qquad I_{\mathrm{CS}}[A] = \frac{k}{4\pi} \int \mathrm{Tr}\Big(A \wedge \mathrm{d}A + \tfrac{2}{3} A \wedge A \wedge A\Big), \qquad k = \frac{\ell}{4G}.} \tag{59}$$

(The level k depends on the normalisation of Tr; here Tr is the trace in the fundamental 2×2 representation of $\mathfrak{sl}(2,\mathbb{R})$. Conventions in the literature differ by factors of 2; the invariant statement is $c = 6k$.)

Three immediate payoffs:

(i) **The equations of motion are $F = \bar{F} = 0$.** All the content of §2.2 (lecture 1) — "every solution is locally AdS_3, and the physics is holonomy" — is now the statement that a flat connection on $\mathcal{M}$ is characterised by its holonomies $\mathrm{Hol}_\gamma(A) = \mathcal{P} e^{\oint_\gamma A}$ around the generators of $\pi_1(\mathcal{M})$. Comparing with §2.3 (lecture 1): the pair $\big(\mathrm{Hol}(A), \mathrm{Hol}(\bar{A})\big)$ around the φ-cycle *is* the pair (g_L, g_R) of (32) (lecture 1).

(ii) **Chirality is manifest.** A and $\bar{A}$ are completely decoupled, which is the bulk origin of the factorisation of the dual CFT into holomorphic and antiholomorphic sectors.

(iii) **Generalisation is trivial.** Replacing $\mathfrak{sl}(2,\mathbb{R}) \to \mathfrak{sl}(N,\mathbb{R})$ gives higher-spin gravity with W_N asymptotic symmetry; replacing it by $\mathfrak{isl}(2,\mathbb{R}) = \mathfrak{iso}(2,1)$ gives the flat-space ($\Lambda = 0$) theory and BMS_3.

Remark 5 (Range of validity)**.** The equivalence (59) is *not* exact as a quantum statement, and not even complete classically. Chern–Simons theory happily allows degenerate e^a_μ (indeed $A = \bar{A}$, i.e. $e = 0$, is a perfectly good CS configuration), while metric gravity requires an invertible dreibein. The two theories therefore have different configuration spaces, and the path integrals need not agree. Witten returns to this at length in [4].

4.2 The Bañados gauge and the central charge

Choose the radial gauge $A = b^{-1}(\mathrm{d} + a)b$ with $b = e^{\log \rho\, L_0}$, so that all ρ-dependence is a similarity transformation and $a = a_+ \mathrm{d}x^+$ carries the physics. The Brown–Henneaux boundary conditions of §3.2 become the *Drinfeld–Sokolov* (highest-weight) form

$$a = \Big(L_1 - \mathcal{L}(x^+)\, L_{-1}\Big)\mathrm{d}x^+, \qquad \bar{a} = -\Big(L_{-1} - \bar{\mathcal{L}}(x^-)\, L_1\Big)\mathrm{d}x^-, \tag{60}$$

with $[L_m, L_n] = (m-n)L_{m+n}$ the $\mathfrak{sl}(2,\mathbb{R})$ generators. Unpacking (57) reproduces exactly the Bañados metric (33) with the same $\mathcal{L}, \bar{\mathcal{L}}$ — the notation is not a coincidence. The residual gauge transformations preserving (60) are parametrised by a single function $\varepsilon(x^+)$ and act on $\mathcal{L}$ by (42); the boundary term in the CS variational principle gives the charge (48), and the central charge comes out as

$$\boxed{c = 6k = 6 \cdot \frac{\ell}{4G} = \frac{3\ell}{2G}} \tag{61}$$

in agreement with §3. [arithmetic checked; the identification $c = 6k$ is the standard $\mathfrak{sl}(2)$ Drinfeld–Sokolov result, quoted from [16, 19].]

4.3 Smoothness as a holonomy condition

The CS language turns "is this black hole geometry smooth?" into an algebraic question. Consider the Euclidean section of BTZ: it is a solid torus, in which the *thermal* cycle $(t_E, \varphi) \to (t_E + \beta, \varphi + \beta\Omega_H)$ is *contractible*. A flat connection extends smoothly over a disc only if its holonomy around the contracting cycle is trivial in the gauge group, which for $\mathrm{SL}(2,\mathbb{R})$ (or its double cover) means

$$\mathrm{Hol}_{\mathrm{thermal}}(A) \;\sim\; -\mathbf{1}, \qquad \text{i.e.} \qquad \text{eigenvalues of } \oint_{\mathrm{thermal}} a \;=\; \pm 2\pi i. \tag{62}$$

Imposing (62) on (60) reproduces exactly the smoothness relation $\beta = 2\pi\ell^2 r_+/(r_+^2 - r_-^2)$ derived from the conical-deficit argument in §3.5 — no metric required. This reformulation is what generalises to higher-spin black holes, where there is no metric-based notion of a horizon and (62) becomes the *definition* of a black hole.

Exercise 8. Verify that (57) turns the first-order action into (59) with $k = \ell/4G$, and that $F = \bar{F} = 0$ is equivalent to zero torsion plus constant curvature $-1/\ell^2$.

Exercise 9. Starting from (60) with constant $\mathcal{L}, \bar{\mathcal{L}}$, reconstruct e^a and ω^a and check that the metric is (33). Then impose (62) and recover the BTZ temperature.

5 Liouville theory and the boundary gravitons

We have found an infinite-dimensional symmetry acting on a phase space labelled by two functions $\mathcal{L}(x^+), \bar{\mathcal{L}}(x^-)$. That phase space is not empty — and since the bulk has no local degrees of freedom, whatever lives in it must live on the boundary. These are the **boundary gravitons**: pure-gauge-looking diffeomorphisms that are not actually gauge, because they carry non-vanishing charge (§3.1). What is their action?

5.1 The reduction: Chern–Simons → WZW → Liouville

The derivation, due to Coussaert, Henneaux and van Driel [16], is a two-step Hamiltonian reduction:

Step 1. CS → chiral WZW. Chern–Simons theory on a manifold with boundary is not a topological theory with no states: the gauge transformations that do not vanish at the boundary become physical. Integrating out the bulk leaves a chiral $\mathrm{SL}(2,\mathbb{R})$ Wess–Zumino–Witten model at level $k = \ell/4G$ on the boundary cylinder, one for each of A and $\bar{A}$. This step uses only the presence of a boundary, not the specific boundary conditions.

Step 2. WZW → Liouville. The Brown–Henneaux conditions are *additional* constraints: in the form (60) they set the coefficient of L_1 in a_+ to the fixed constant 1. These are first-class constraints of Drinfeld–Sokolov type, and imposing them on the WZW model is precisely the Hamiltonian reduction $\mathrm{SL}(2,\mathbb{R})_k$ WZW $\longrightarrow$ Liouville.

The outcome:

$3d$ Einstein gravity, $\Lambda < 0$, Brown–Henneaux boundary conditions	$\Longleftrightarrow$	Liouville theory on the boundary cylinder, $c = \dfrac{3\ell}{2G}$

(63)

with action, in one common normalisation,

$$S_{\rm L} = \frac{\ell}{16\pi G}\int \mathrm{d}x^+\mathrm{d}x^- \,\Big[\partial_+\phi\,\partial_-\phi - \,\mu\, e^{\phi}\Big], \tag{64}$$

and stress tensor $T = -\frac{1}{4}\big(\partial_+\phi\big)^2 + \frac{1}{2}\partial_+^2\phi$ up to conventions. [The overall normalisation of (64) and the coefficient of the improvement term are convention-dependent and are quoted, not re-derived, from [16]; the central charge $c = 3\ell/2G$ is the convention-independent statement.] The relation to §2.2 is that $\mathcal{L}$ is the Liouville stress tensor and the conical-defect/BTZ classification of §2.2 (lecture 1) becomes the classification of Liouville monodromies.

5.2 Why this is not a proof of AdS_3/CFT — and what it is

Two lessons, both of which we will need repeatedly.

(a) A central charge is not a spectrum. Liouville theory has exactly $c = 3\ell/2G$, yet its spectrum is *continuous*, bounded below by $\Delta \geq (c-1)/24$, with no normalisable $\mathrm{SL}(2,\mathbb{R})$-invariant vacuum. It therefore has neither a state dual to global AdS_3 nor a Cardy growth reproducing $A/4G$ (cf. item (iii) of the aside in §3.6). Whatever the microscopic dual of AdS_3 gravity is, classical Liouville is at best a description of the boundary-graviton sector, not of the full theory.

(b) The modern reformulation: geometric actions on Virasoro orbits. The invariant content of the reduction is that the gravitational phase space is a set of *coadjoint orbits of the Virasoro group*:

$$\{\text{Bañados solutions}\} \;=\; \big(\mathrm{Diff}(S^1)/H\big) \times \big(\mathrm{Diff}(S^1)/\bar{H}\big), \tag{65}$$

where H is the stabiliser of the chosen representative $\mathcal{L}_0$: $H = \mathrm{PSL}(2,\mathbb{R})$ for global AdS_3 ($\mathcal{L}_0 = -\frac{1}{4}$), and $H = U(1)$ for a BTZ black hole ($\mathcal{L}_0 > 0$). The dynamics on each orbit is the Alekseev–Shatashvili geometric action; for the BTZ orbit it reduces to a Schwarzian-like theory, and this is what computes the one-loop "boundary graviton" partition function

$$Z^{\mathrm{AdS}_3}_{1-\mathrm{loop}} = \prod_{n=2}^{\infty}\frac{1}{|1-q^n|^2}, \qquad q = e^{2\pi i\tau}, \tag{66}$$

of Maloney–Witten [5]: the product starts at $n = 2$ precisely because $n = 0, \pm 1$ are the $\mathrm{SL}(2,\mathbb{R})$ isometries, which are *not* boundary gravitons. The modern treatments are Cotler–Jensen [6] and Barnich et al.

Exercise 10. Identify the stabiliser H of $\mathcal{L}_0 = -\frac{1}{4}$ inside $\mathrm{Diff}(S^1)$ using (42), and show it is $\mathrm{PSL}(2,\mathbb{R})$. Repeat for $\mathcal{L}_0 = \mathrm{const} > 0$ and show the stabiliser is only $U(1)$. Conclude that the black hole orbit is "larger" than the vacuum orbit, and count the boundary gravitons in each case.

Where this goes next

A running list of candidate topics for Weeks 2–12 — higher-spin Chern–Simons, Euclidean quantum gravity and the $\mathrm{SL}(2,\mathbb{Z})$ sum, holographic renormalisation, entanglement entropy, higher-genus quotients and wormholes, warped AdS_3, supersymmetric AdS_3, flat-space holography, and the ensemble-averaging question — is kept in the companion document `topics.tex`.

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